

## Problem

Design a circuit that uses a d'Arsonval meter (with an internal resistance of  $2\text{ k}\Omega$  that requires a current of  $5\text{ mA}$  to cause the meter to deflect full scale) to build a voltmeter to read values of voltages up to  $100\text{ volts}$ .

## Step-by-step solution

## Step 1 of 3

Consider the following data:

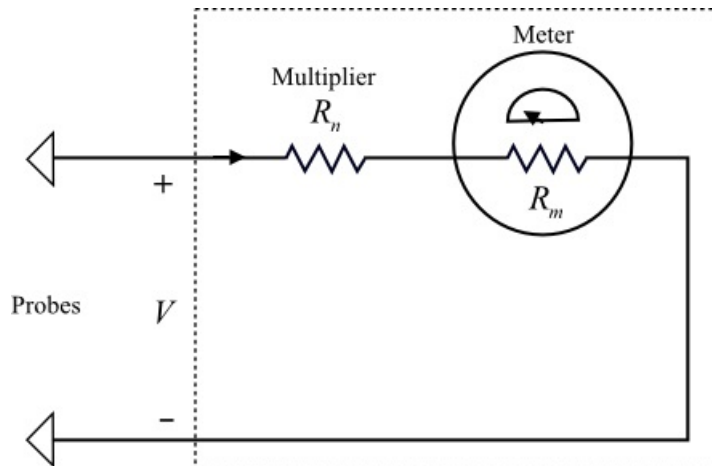
The internal resistance of a d'Arsonval meter,  $R_m = 2\text{ k}\Omega$

The full scale current,  $I_{fs} = 5\text{ mA}$

The full scale voltage,  $V_{fs} = 100\text{ V}$

## Step 2 of 3

Draw the circuit diagram according to the data.



## Step 3 of 3

Calculate the series resistance needed to measure  $100\text{ V}$  full scale deflection.

$$\begin{aligned}
 R_n &= \frac{V_{fs}}{I_{fs}} - R_m \\
 &= \frac{100}{5 \times 10^{-3}} - 2 \times 10^3 \\
 &= (20 \times 10^3) - (2 \times 10^3) \\
 &= 18\text{ k}\Omega
 \end{aligned}$$

Therefore, the value of the series resistance needed to measure the  $100\text{ V}$  full scale is

$$18\text{ k}\Omega$$